

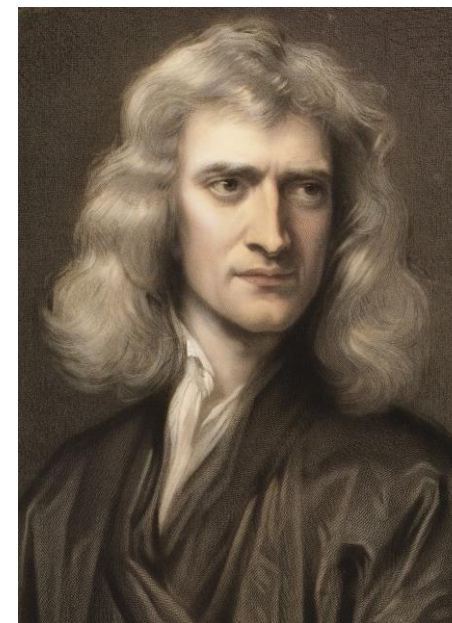
Math Primer

(10. Partial Derivatives)

Contents



- ▶ Functions of several variables
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- ▶ Second order and mixed partials
- ▶ Total Derivative
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- ▶ Chain Rule
- ▶ Gradient
- ▶ Jacobian
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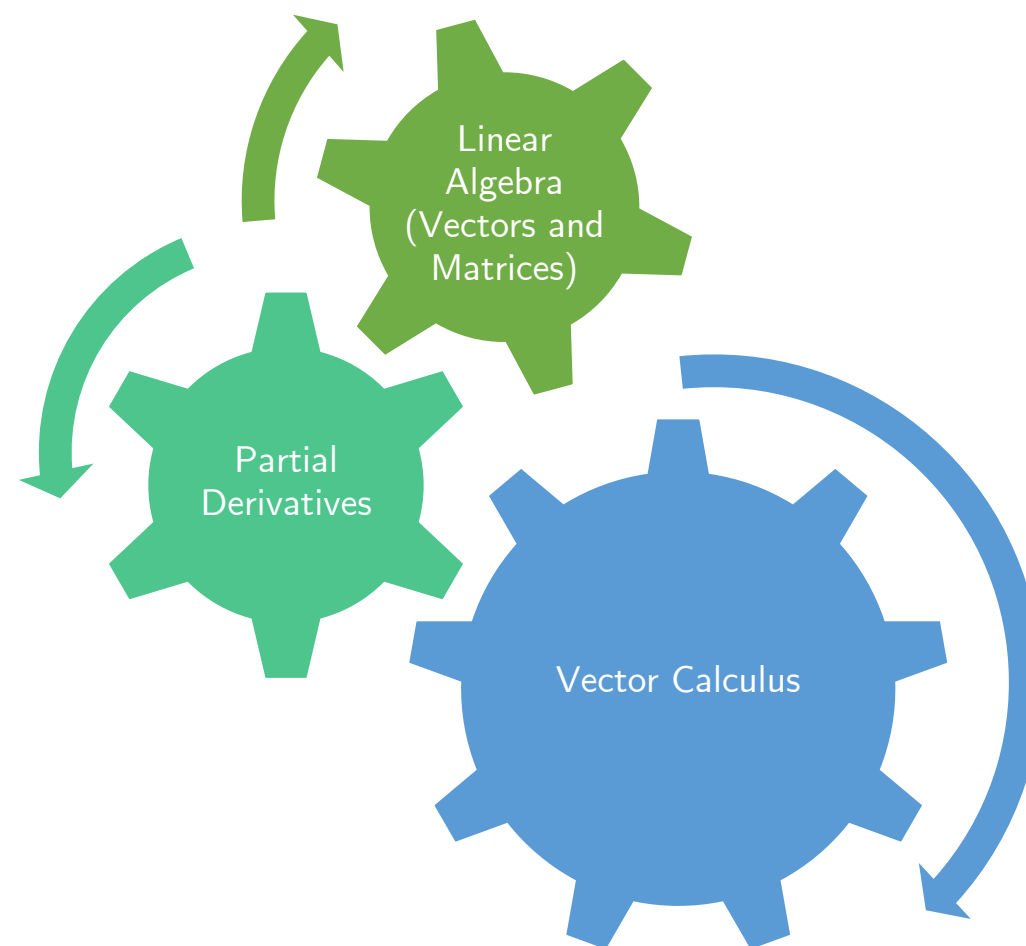


Isaac Newton
(1643-1727)



G.W. Leibniz
(1646-1716)

The Multivariable Calculus Landscape



Functions of several variables

- Often, we come across functions which depend on multiple input variables. We call those multivariable functions (function of several variables). A bivariate function for example is

$$\underline{z} = f(x, y)$$

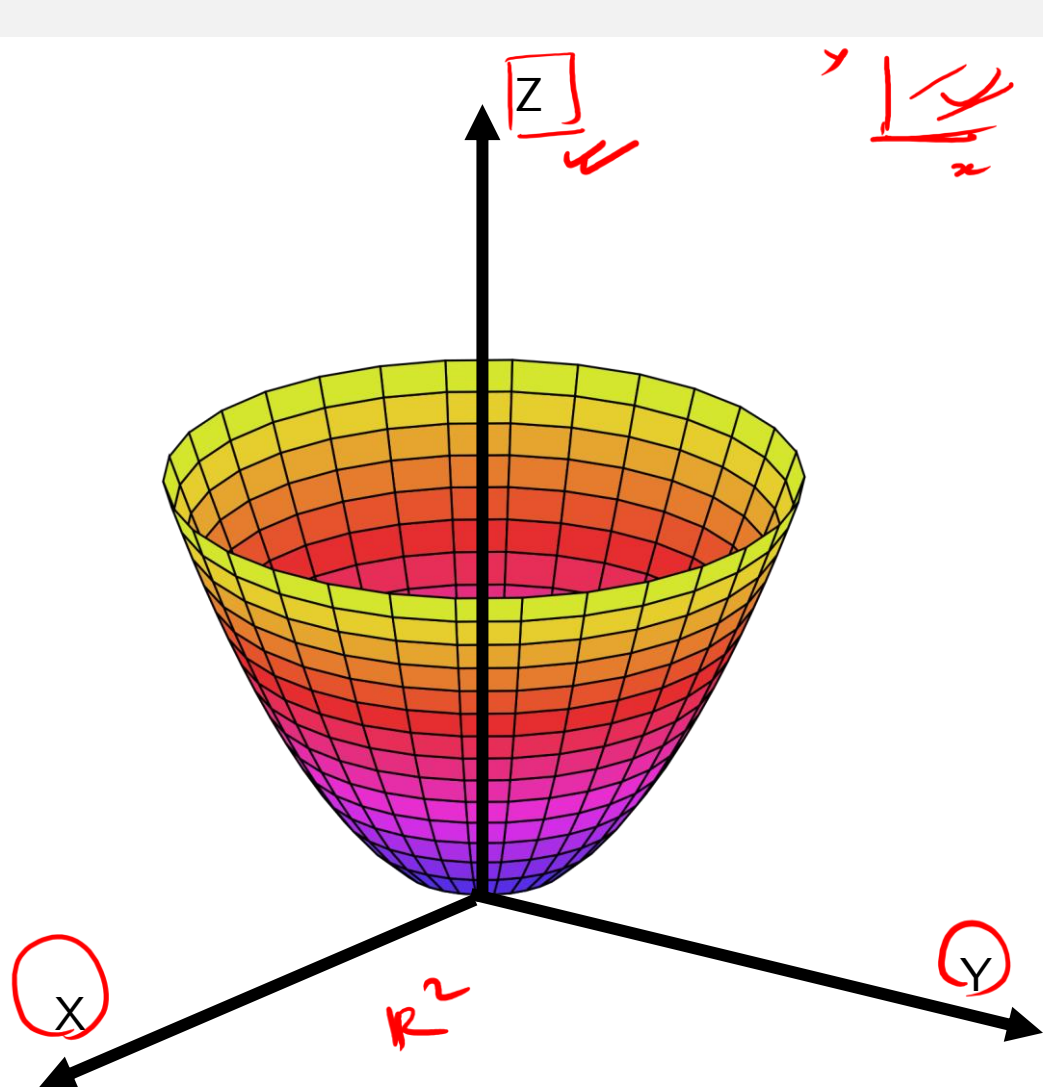
- Other examples include

Utility function : $\underline{u}(x, y) = x^{\frac{1}{2}}y^{\frac{1}{2}}$ $\sqrt{xy}$

Option Price : $\underline{V}(S, K, r, \tau, \sigma) = SN(d_1) - Ke^{-r\tau}N(d_2)$, $d_1 = \frac{\ln(\frac{S}{K}) + (r + 0.5\sigma^2)T}{\sigma\sqrt{T}}$, $d_2 = d_1 - \sigma\sqrt{T}$

- In this chapter, we will talk about partial derivatives which are used for
- I. Measuring the rate of change of output to input i.e., sensitivity or Greeks in finance,
 - II. Approximation of change in a function (multivariate Taylor Series) ,
 - III. Optimization (Maxima, Minima)

Visualizing in 3D ...



$$y = x^2$$

$$\underline{z} = \underline{f}(\underline{x}, \underline{y}) = \underline{x}^2 + \underline{y}^2$$

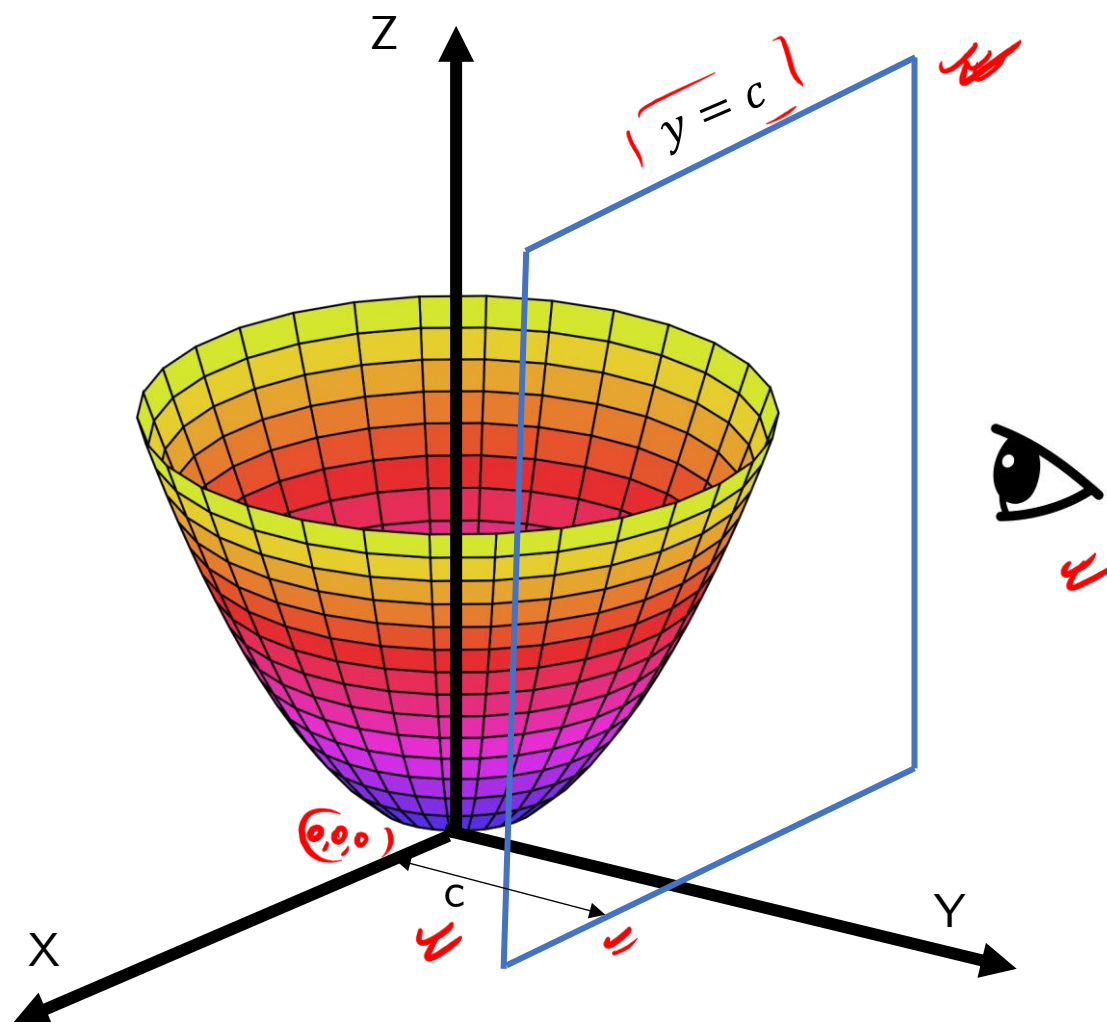
Output/Dependent variable Input/Independent variables

$$\text{Domain : } (x, y) \in \underline{\mathbb{R}^2}$$

$$\text{Range : } z \in \underline{\mathbb{R}^+}$$

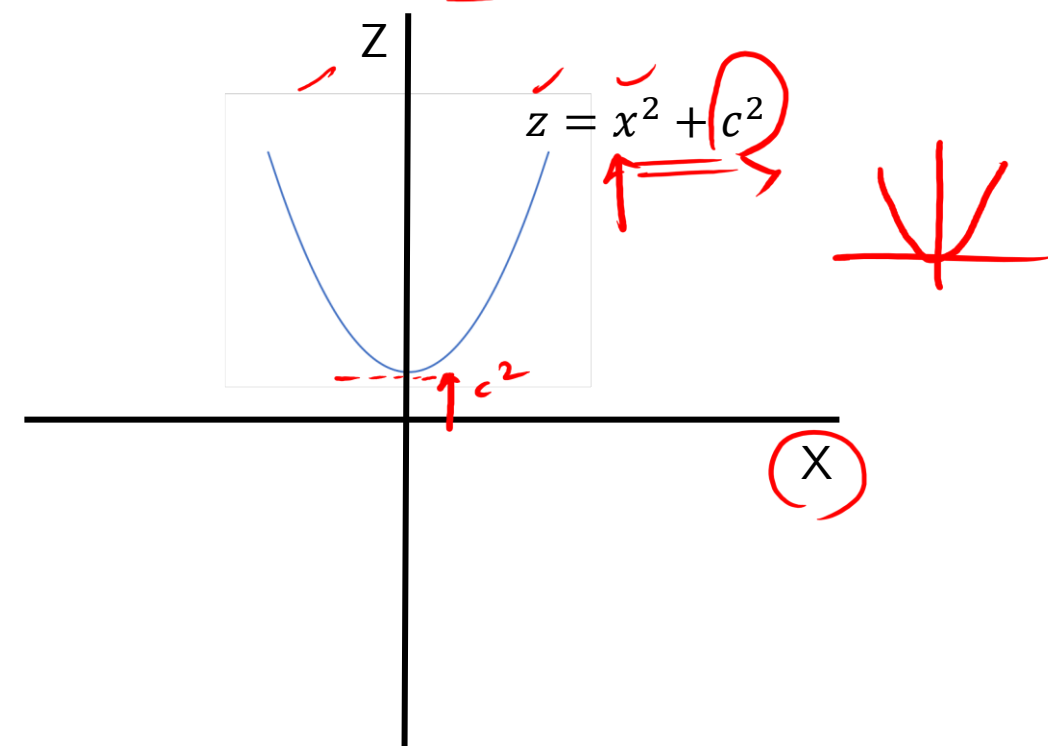
$$\underline{f} : \underline{\mathbb{R}^2} \rightarrow \underline{\mathbb{R}^+}$$

Intersection of a plane and a surface

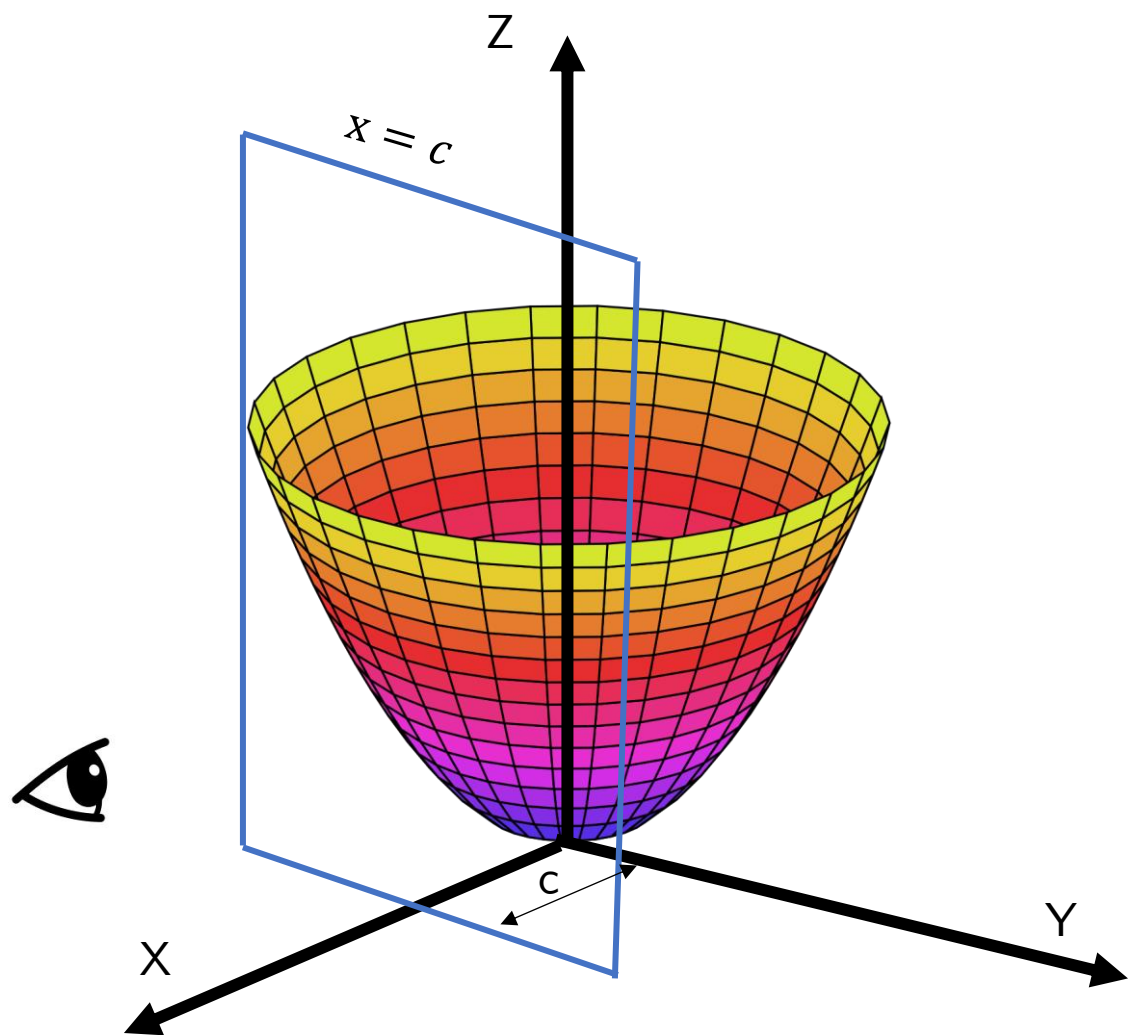


$$z = f(x, y) = x^2 + y^2$$

What is the shape of the section cut by the plane $y = c$?

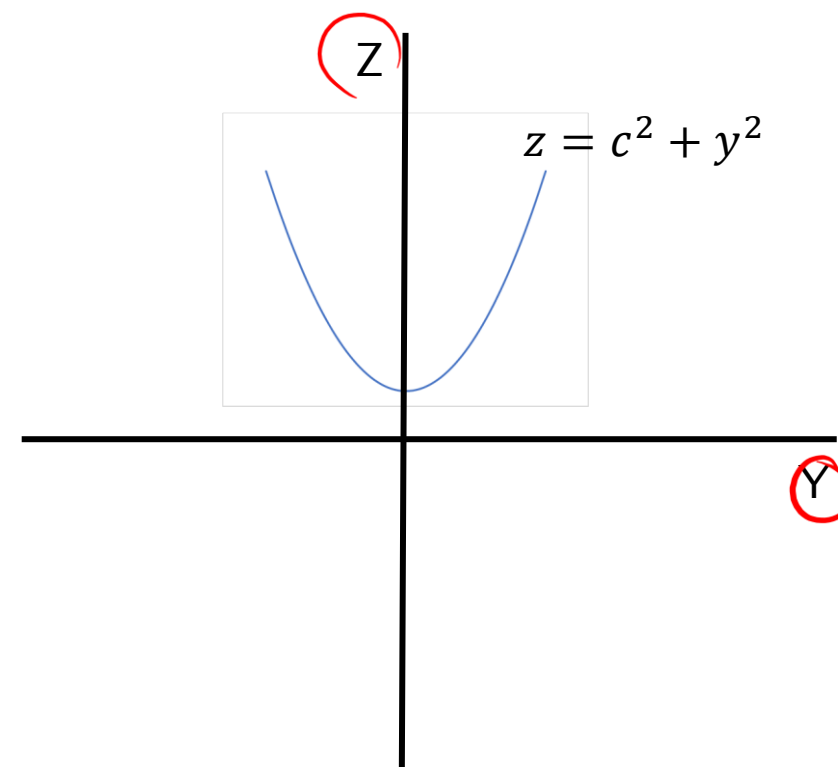


Intersection of a plane and a surface

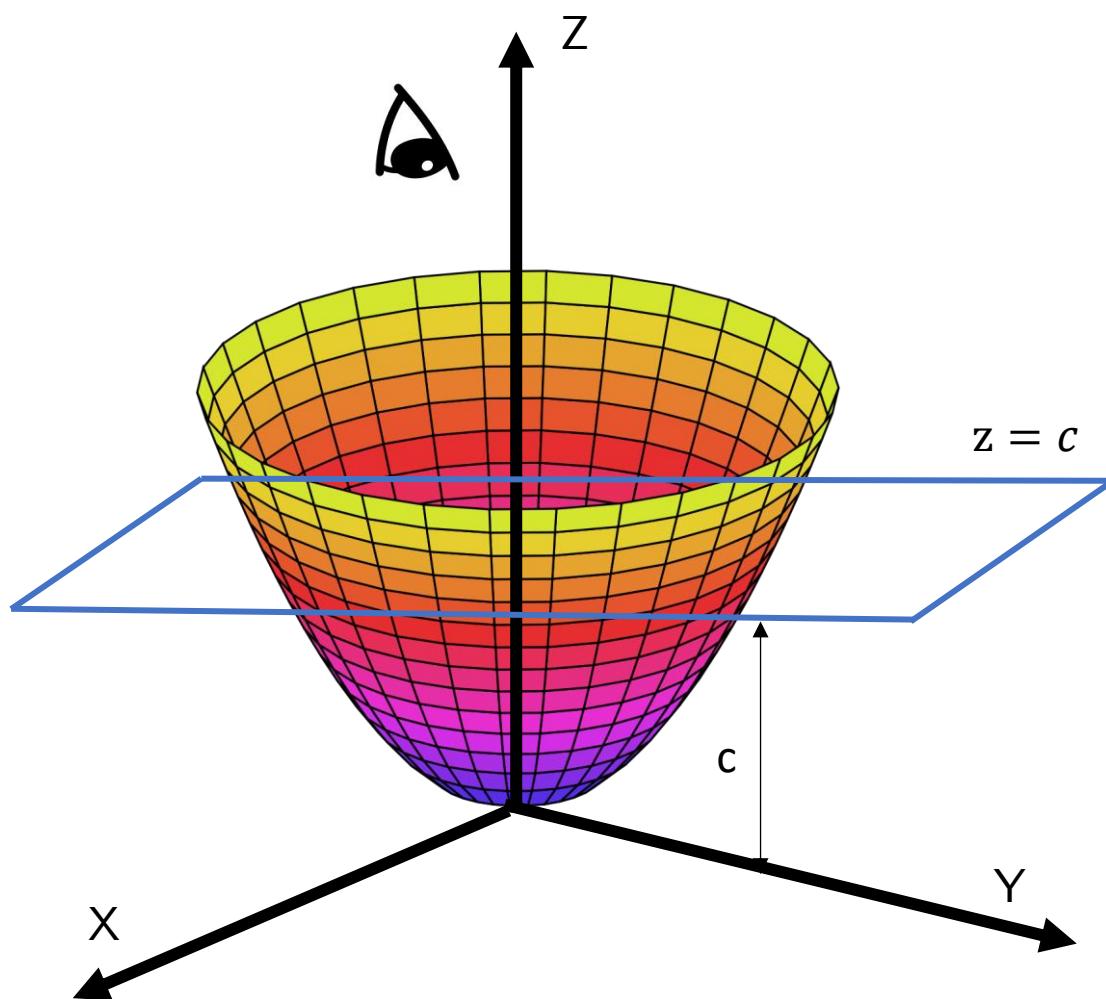


$$z = f(x, y) = x^2 + y^2$$

What is the shape of the section cut by the plane $x = c$?

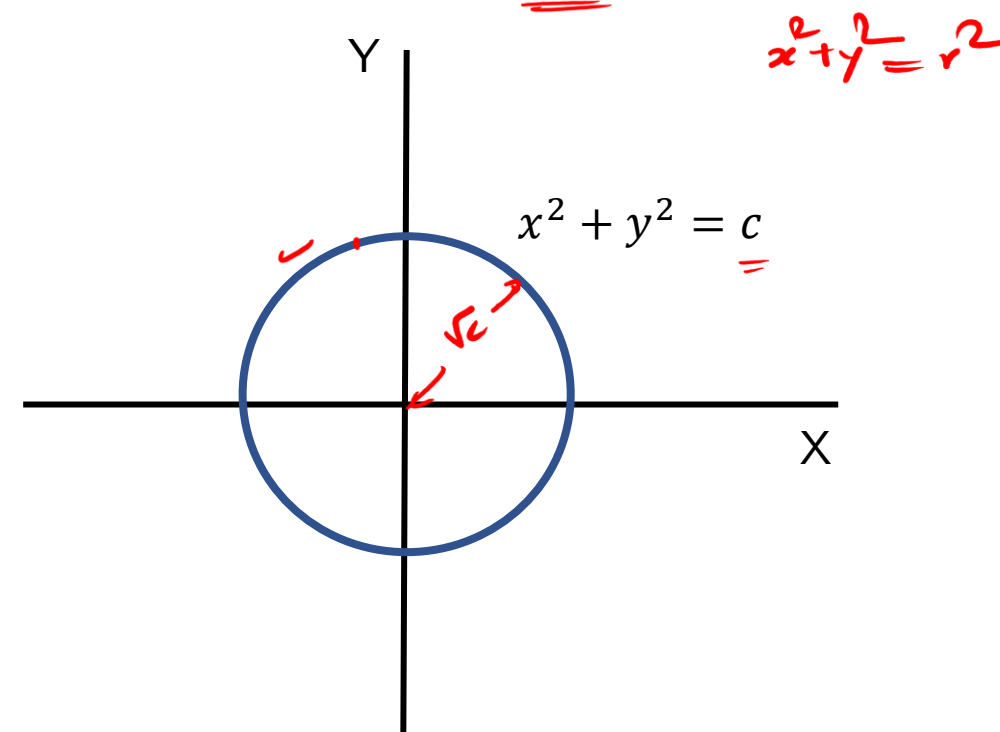


Level Curve

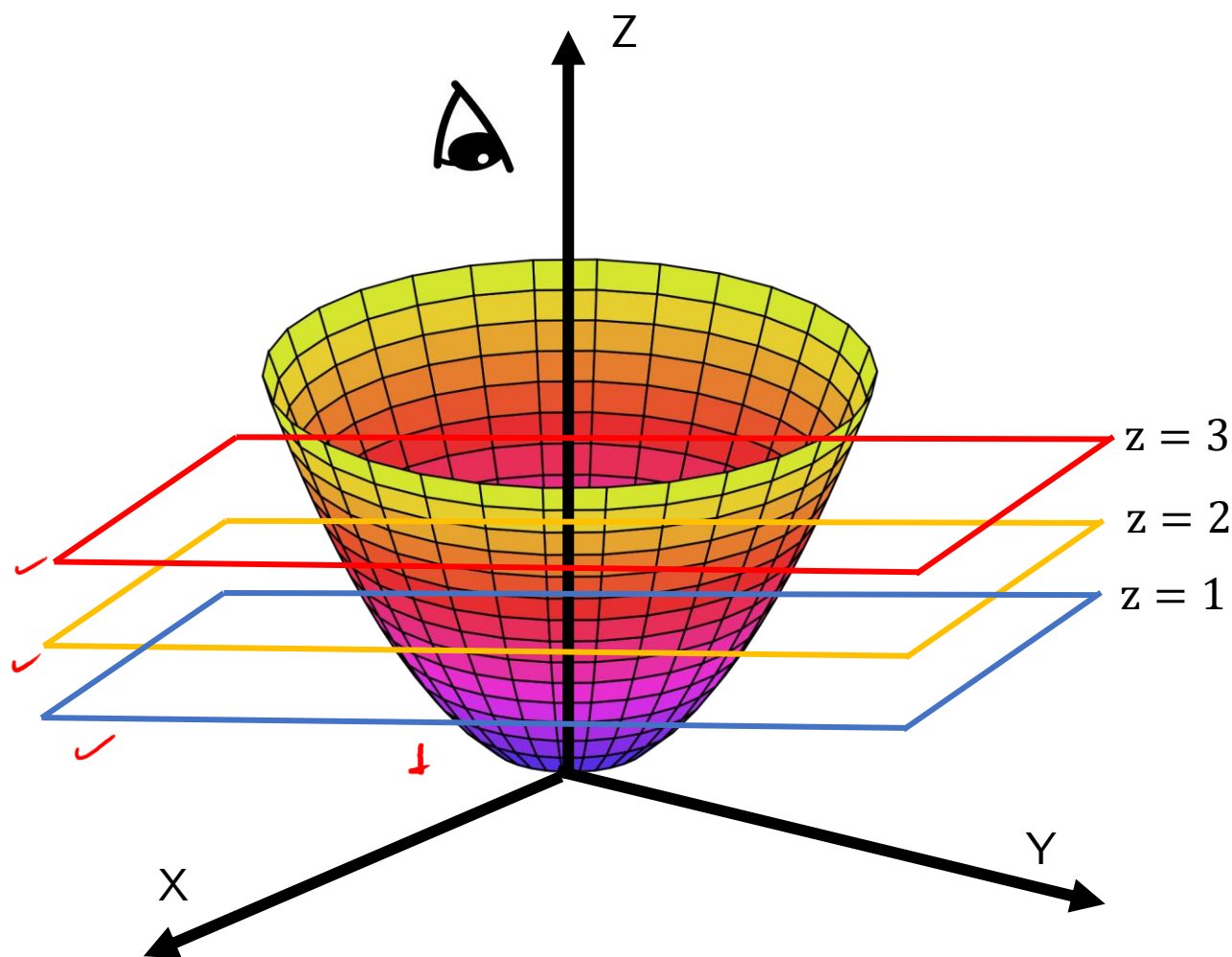


$$z = f(x, y) = \underline{x^2 + y^2}$$

- ▶ If we fix $z = c$, we get a circle of points which share the same height $= c$.
- ▶ This is called as a level curve

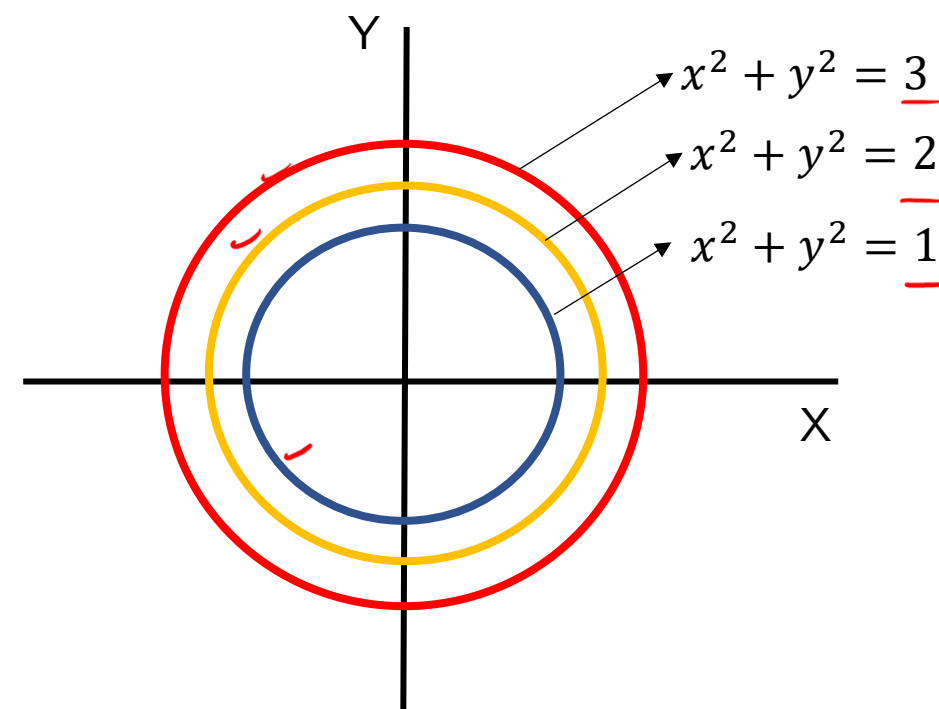


Contour Plot



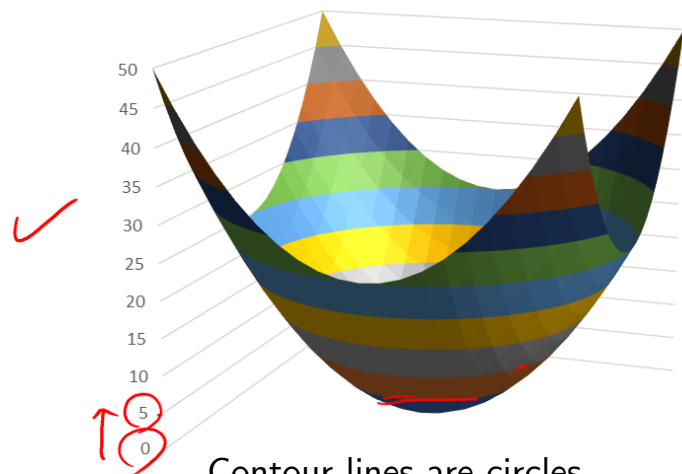
$$z = f(x, y) = x^2 + y^2$$

- By changing c , we can generate many level curves. The set is called **contour plot**.

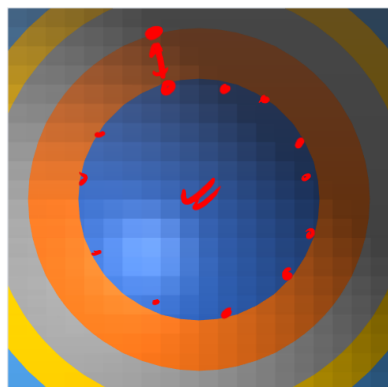


Contour Plots - Examples

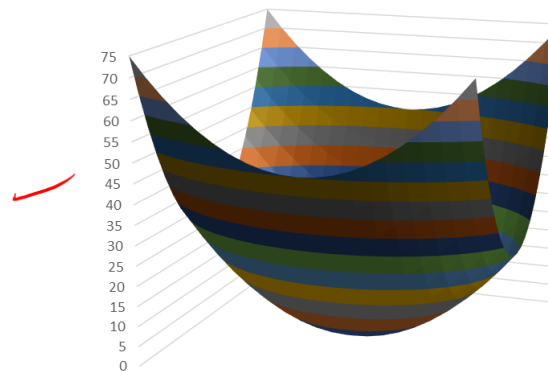
$$z = f(x, y) = x^2 + y^2$$



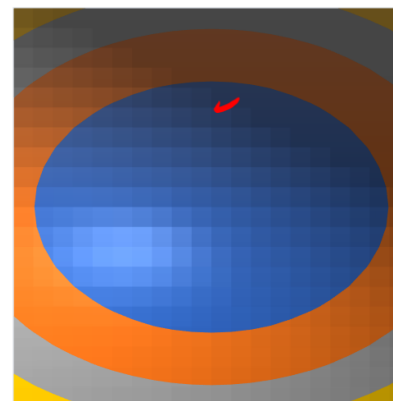
Contour lines are circles



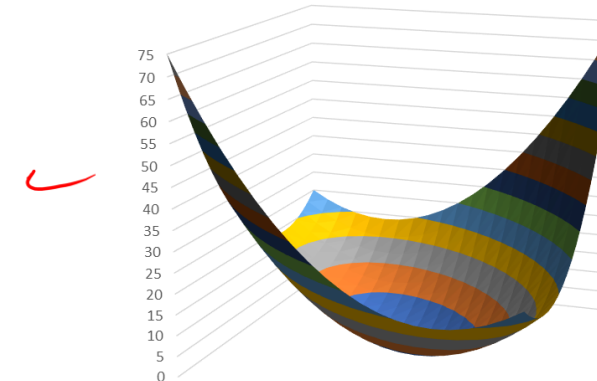
$$z = f(x, y) = x^2 + 2y^2$$



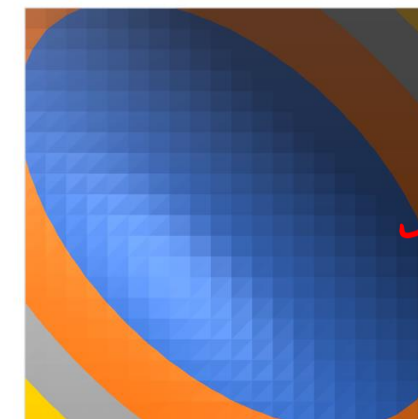
Contour lines are ellipses



$$z = f(x, y) = x^2 + xy + y^2$$



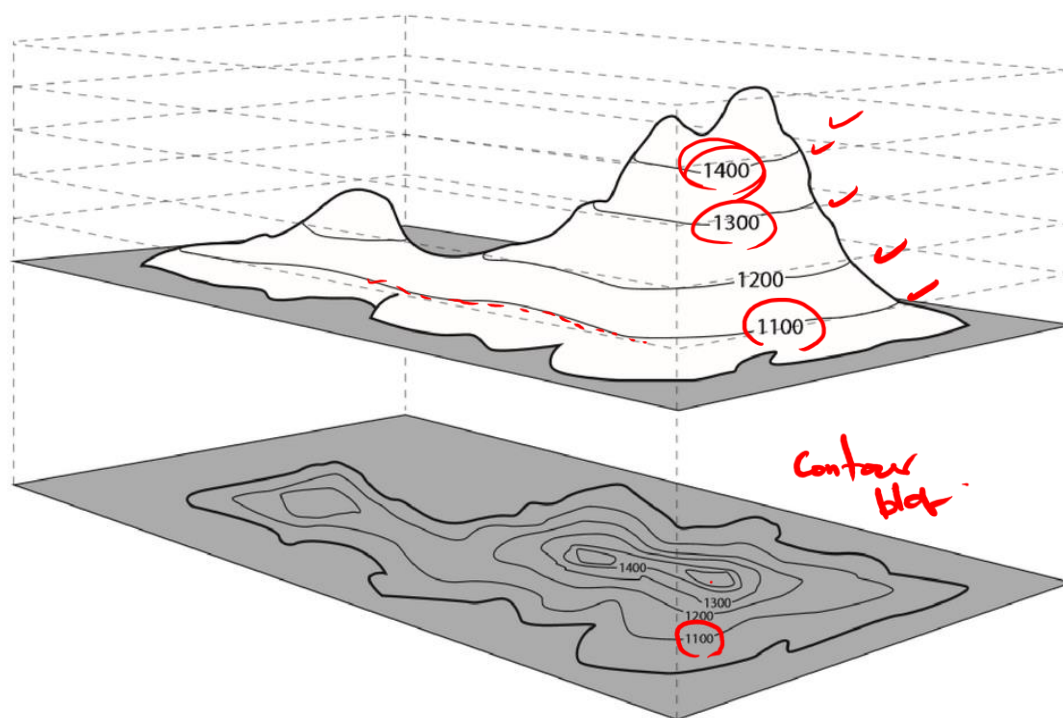
Contour lines are rotated ellipses



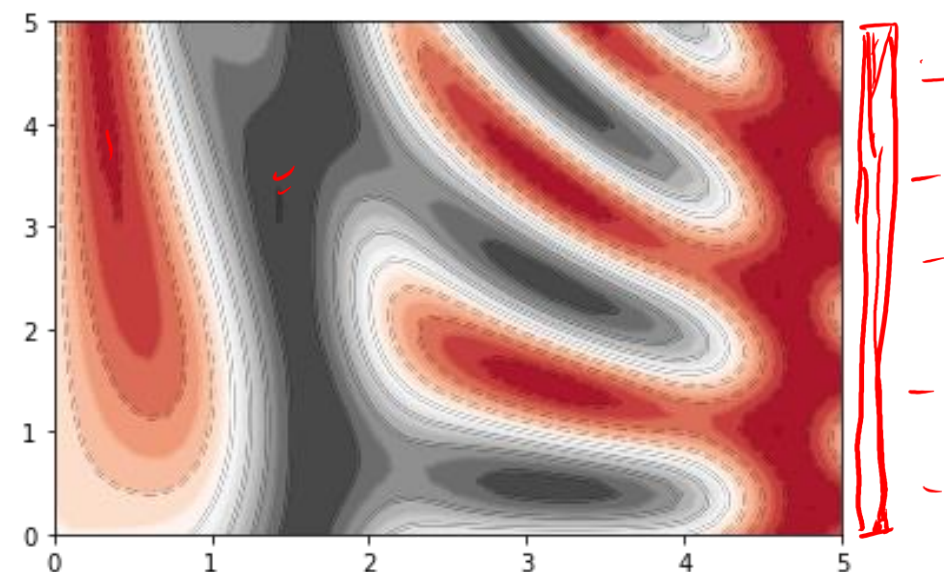
Topography and Heat Maps

- ▶ Topographical plots and heatmaps are common visualization tools to gauge the elevation

Topographical map

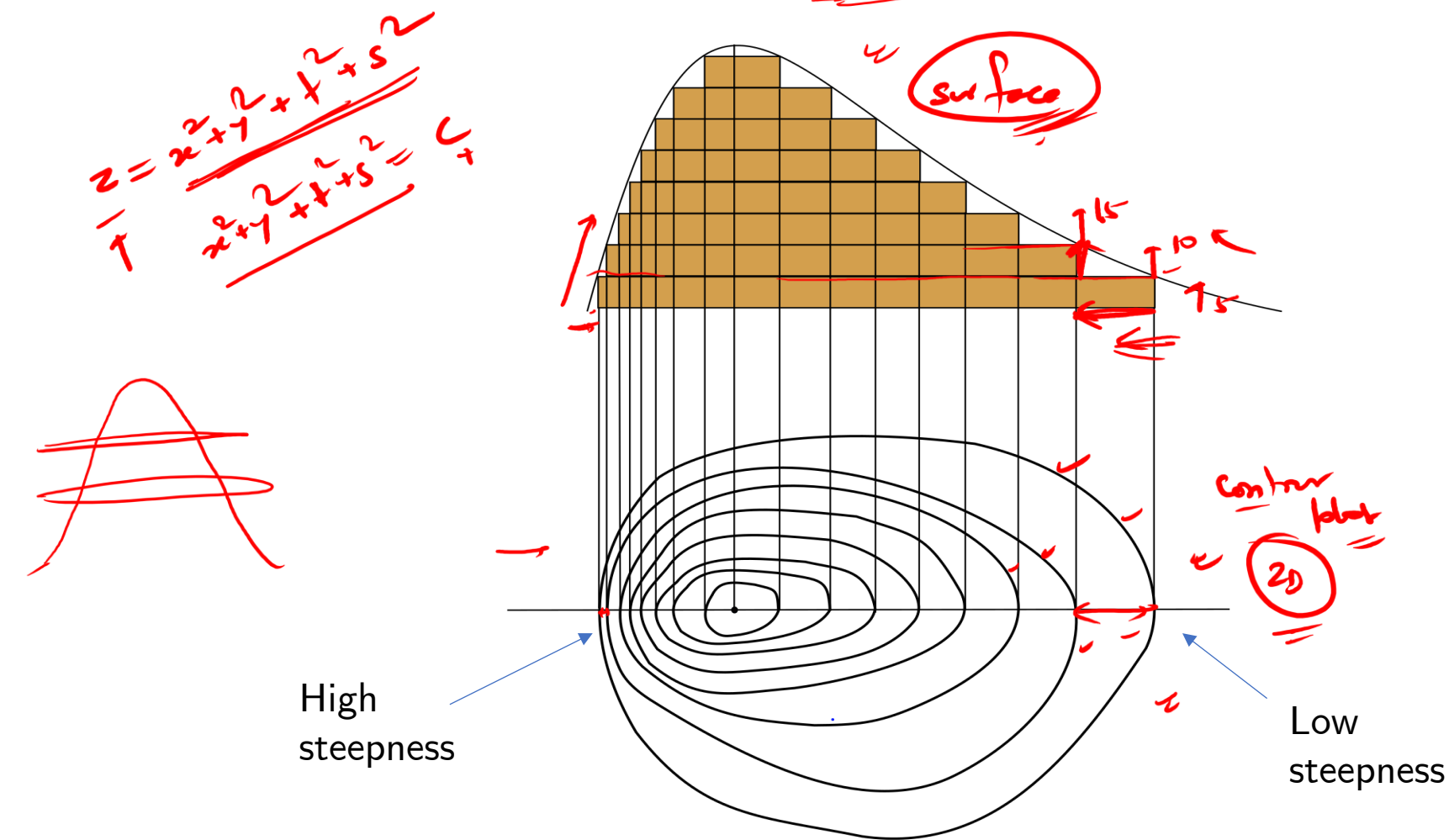


Heat Maps



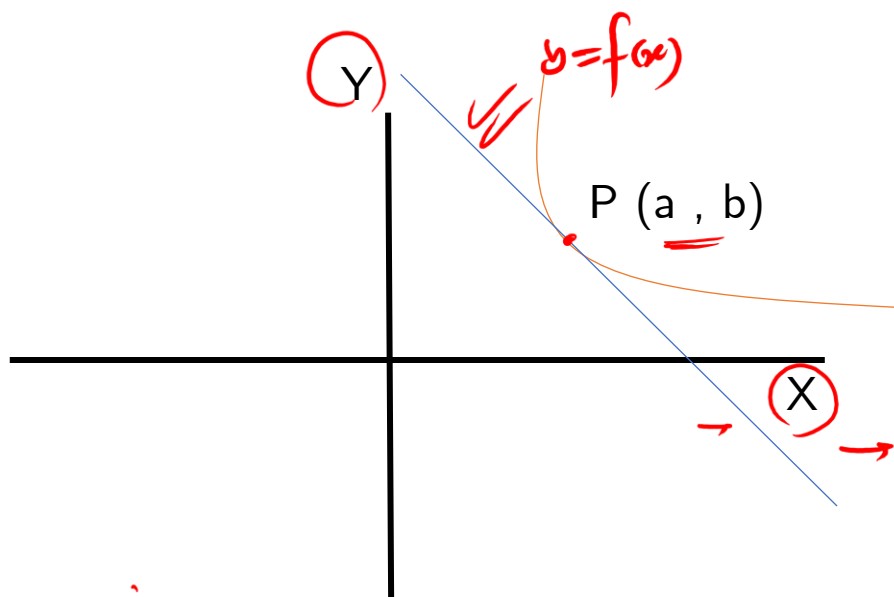
What does contour plot tell us ?

- ▶ Contour plots can give an idea of steepness.

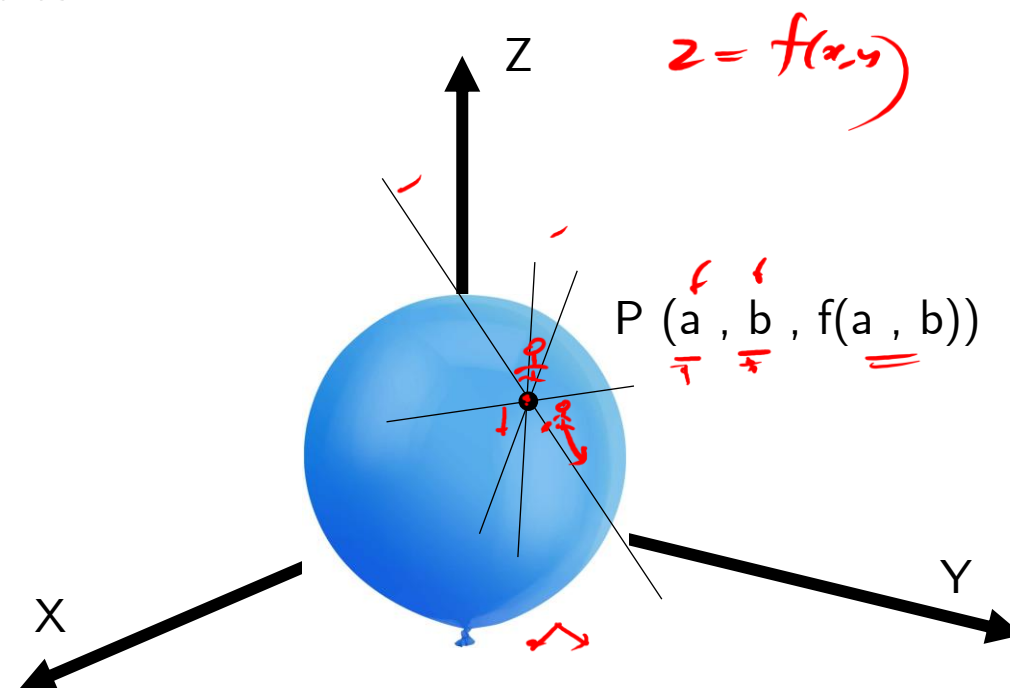


Derivatives in the presence of multiple variables

- For a curve in 2D, the derivative at a point is the slope of the tangent to the curve which is unique. But in case of multiple variables, the derivative at a point can exist in infinite directions (infinite tangents exist). Therefore, we need to specify a direction when we talk about differentiating in the presence of multiple input variables.



Derivative is the slope of the tangent to the curve



So many possible tangents !!

Derivative along $x := f_x$

- Consider a bivariate function $z = f(x, y)$. We can calculate the derivative of z w.r.t x without changing y . It's called as **partial derivative** along x and is denoted as

$$f_x = \frac{\partial f}{\partial x} \text{ (first order partial derivative w.r.t } x \text{)}$$

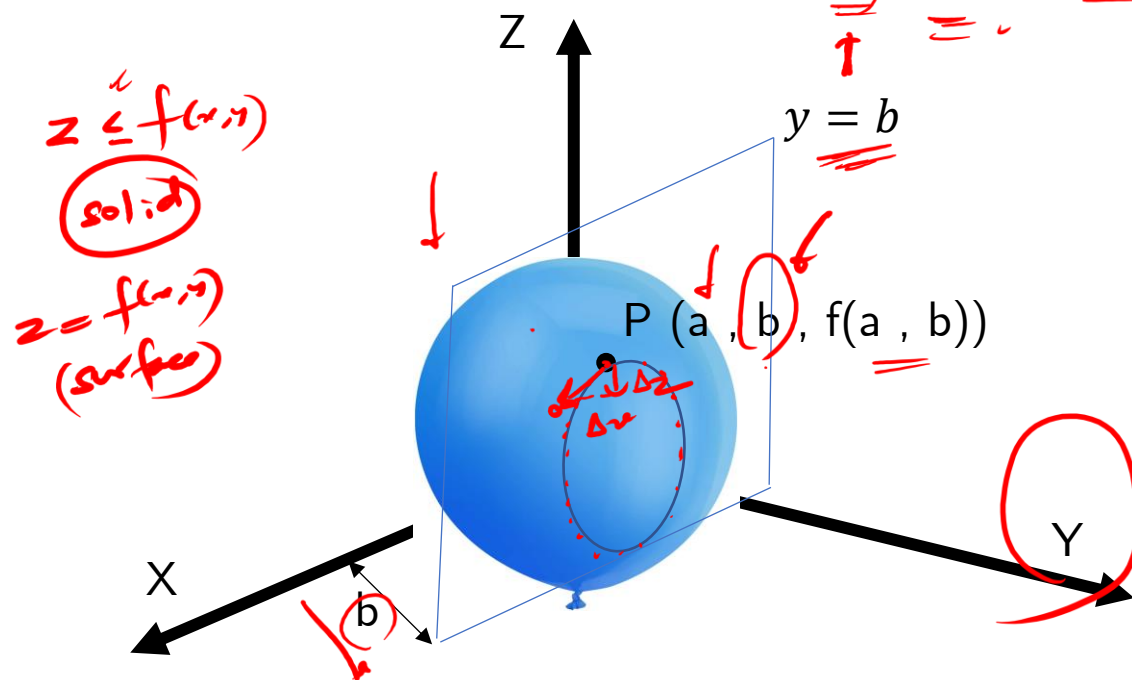
(d)

$$y = x^2$$

$$2x$$

$$x = 5$$

$$10$$



Derivative of z along x is ratio of change of z when x changes by a small amount keeping y constant

The curve generated by intersection of the surface and the plane $y = b$, x is the only independent variable, and y is kept constant at b

$$z = f(x, y) \rightarrow \text{surface}$$

$$z = f(x, b) \rightarrow \text{curve}$$

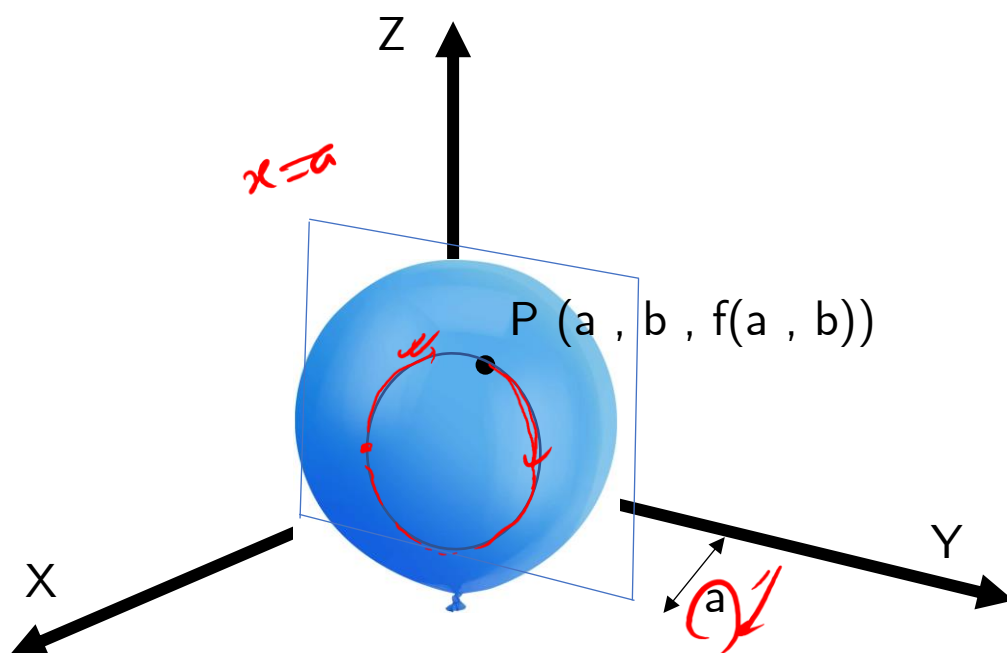
$$\frac{\partial f}{\partial x} = f_x = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$\left(\frac{\partial f}{\partial x} \right)_{x=a, y=b} = \lim_{\Delta x \rightarrow 0} \frac{f(a + \Delta x, b) - f(a, b)}{\Delta x}$$

Derivative along $y := f_y$

- We can similarly calculate partial derivative along y

$$\underline{f_y} = \underline{\frac{\partial f}{\partial y}} \text{ (first order partial derivative w.r.t } y \text{)}$$



Derivative of z along y is ratio of change of z when y changes by a small amount keeping x constant

The curve generated by intersection of the surface and the plane $x = a$, y is the only independent variable, and x is kept constant at a

$z = f(x, y)$ → surface

$z = f(a, y)$ → curve

$$\underline{\frac{\partial f}{\partial y}} = f_y = \lim_{\Delta y \rightarrow 0} \frac{f(x, \underline{y + \Delta y}) - f(x, y)}{\Delta y}$$

$$\left(\frac{\partial f}{\partial y} \right)_{x=a, y=b} = \lim_{\Delta y \rightarrow 0} \frac{f(a, \underline{b + \Delta y}) - f(a, b)}{\Delta y}$$

First-order Partial derivatives

Exercises : Compute the first-order partial derivatives for the following functions

$$1. \quad f(\underline{x}, \underline{y}) = \underline{10} + \underline{2x} + \underline{3y}, \quad f_x = 2, \quad f_y = 3$$

$$2. \quad f(x, y) = 10 + 2x^2 + y^3, \quad f_x = 4x, \quad f_y = 3y^2$$

$$3. \quad f(x, y) = 3x^2y^2, \quad f_x = 6xy^2, \quad f_y = 6x^2y$$

$$4. \quad f(x, y, z) = 10x + 12y + 20z, \quad f_x = 10, \quad f_y = 12, \quad f_z = 20$$

$$5. \quad f(x, y) = \underline{2x^2}(4x + 6y), \quad f_x = 24x^2 + 24xy, \quad f_y = 12x^2$$

$$6. \quad f(x, y) = \frac{x+y}{2y}, \quad f_x = \frac{1}{2y}, \quad f_y = -\frac{x^2}{2y^2}$$

$\frac{x}{2y} + \frac{1}{2}$

Second-order Partial derivatives

► By taking partial derivatives of partial derivatives, we get higher order partials. For example, the following are defined

Second order partials : $f_{xx} = \frac{\partial f_x}{\partial x} = \frac{\partial^2 f}{\partial x^2}$, $f_{yy} = \frac{\partial f_y}{\partial y} = \frac{\partial^2 f}{\partial y^2}$

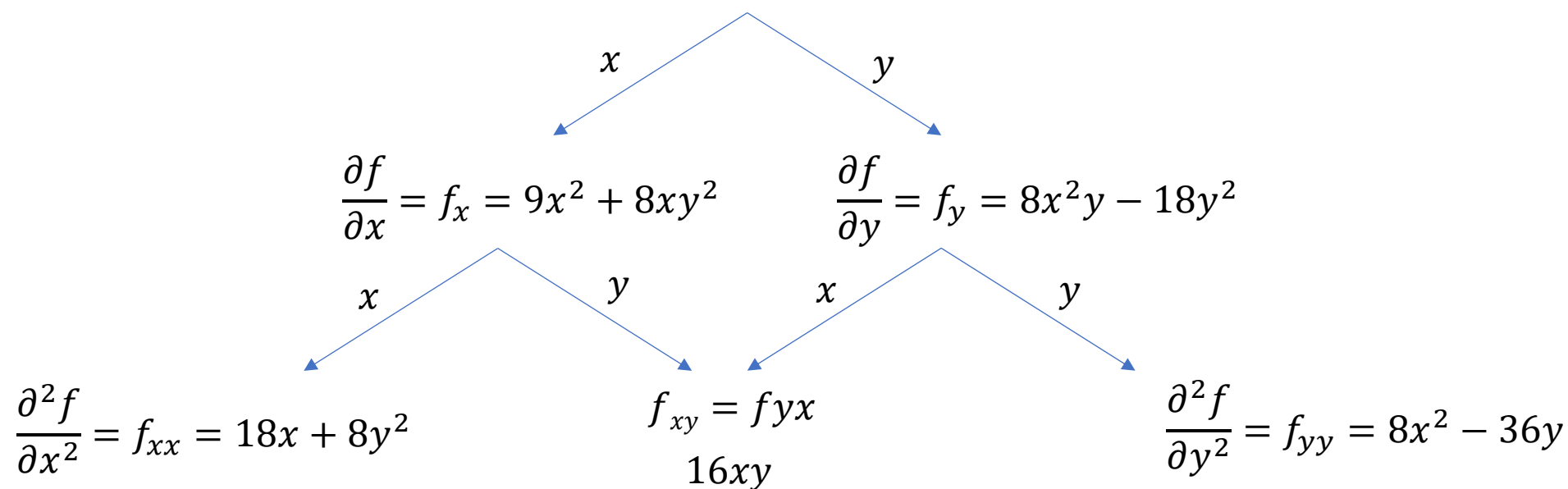
Mixed partials : $f_{xy} = \frac{\partial f_x}{\partial y}$, $f_{yx} = \frac{\partial f_y}{\partial x}$

$$\frac{\partial f_x}{\partial y} = \frac{\partial f_y}{\partial x} \quad (\text{Young's theorem})$$

Exercise : Show that $f_{xy} = f_{yx}$ for $f(x, y) = \underline{3x^3 + 4x^2y^2 - 6y^3}$

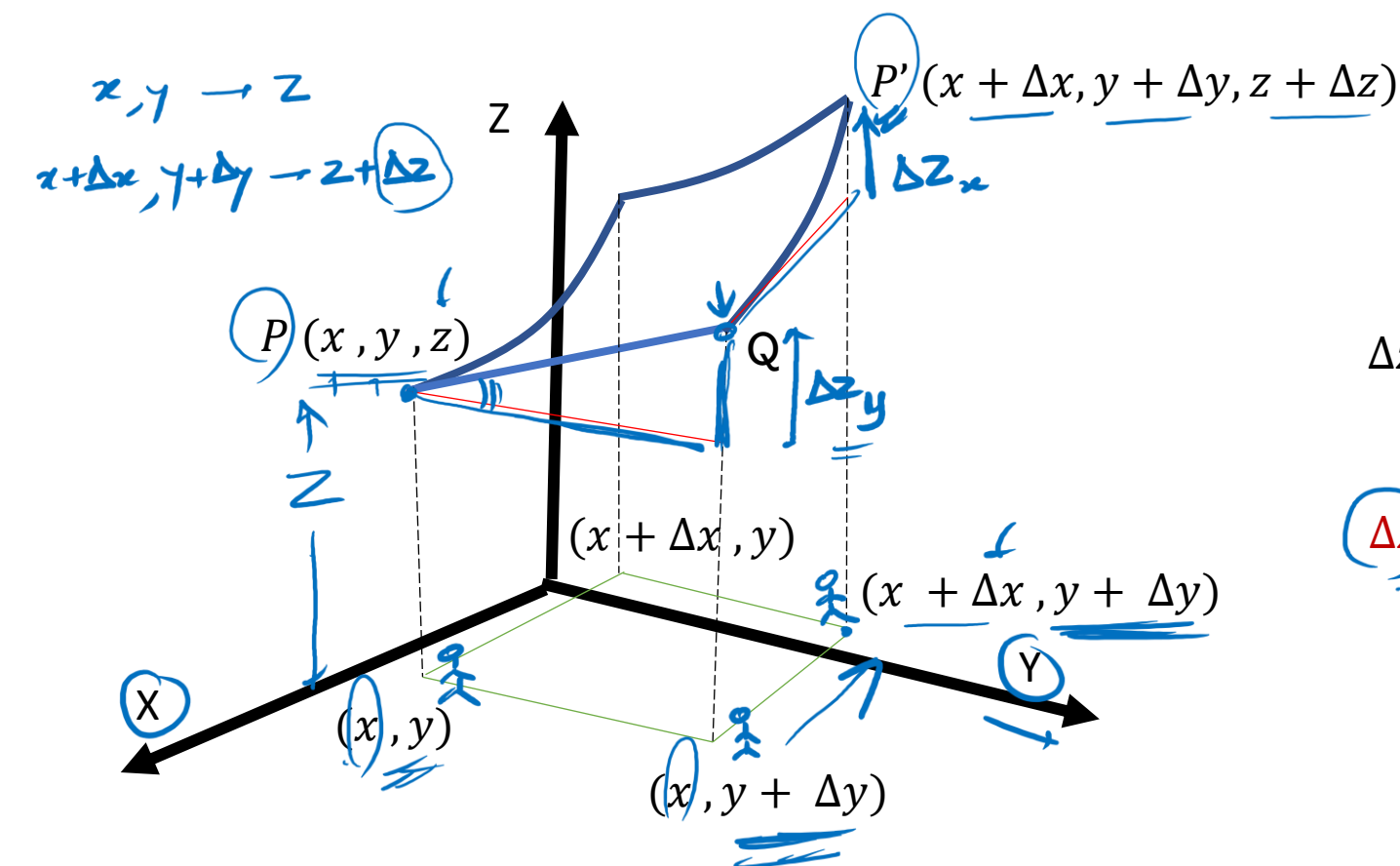
Second-order Partial derivatives

$$z = f(x, y) = 3x^3 + 4x^2y^2 - 6y^3$$



Total Derivative

- ▶ Total differential of a function can be attributed to changes in the inputs through the partial derivatives. We can visualize how much is because of first order change



$$\Delta z = \begin{array}{l} \text{(change because of change in } y \text{)} \\ + \\ \text{(change because of change in } x \text{)} \end{array}$$

$$\Delta z = z_y(\Delta y) + z_x(\Delta x)$$

$$\Delta z = \left(\frac{\partial z}{\partial y} \right) (\Delta y) + \left(\frac{\partial z}{\partial x} \right) (\Delta x)$$

Multivariate Taylor Series

- Now that we have functions of multiple inputs, the Taylor series approximation for the change in the function can be defined up to second order as

$$\Delta f(x, y) \approx f_y \Delta y + f_x \Delta x + \frac{1}{2} f_{yy} (\Delta y)^2 + \frac{1}{2} f_{xx} (\Delta x)^2 + f_{xy} (\Delta x) (\Delta y) + \dots$$

$\frac{1}{2} \frac{\partial^2 f}{\partial x \partial y} (\Delta x) (\Delta y) + \frac{1}{2} \frac{\partial^2 f}{\partial y \partial x} (\Delta x) (\Delta y)$
 $+ \frac{\partial^2 f}{\partial x \partial y} (\Delta x) (\Delta y)$

- Notice there is no $\frac{1}{2}$ in the mixed partial term because it represents the sum of 2 mixed partials which are same

Exercise : Express the total change in $f(x, y) = e^{x+y}$

Chain Rule

- In case of univariate functions, the chain rule we learnt was straightforward. For example, if $y = x^2$ and $x = t^3$, then

①

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = 2x \cdot 3t^2 = 6t^5$$

- In case of functions with several variables, the total derivative is a bit involved. For example, find the following

② $z = x^2 + y^2$
 $= 6^2 + 4^2$
 $\frac{dz}{dt} = 6t^5 + 4t^3$

→ $z = f(x, y) = x^2 + y^2$
 and $y = t^2$ and $x = t^3$, find $\frac{dz}{dt}$

► Using $dz = f_x dx + f_y dy \Rightarrow \frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt} = 2x \cdot 3t^2 + 2y \cdot 2t = 6t^5 + 4t^3$

Chain Rule

- So basically, with multiple input variables, we need to include changes from all possible paths. Let's look at some examples

$$y = x^2 \text{ and } x = t^3$$

$$z = f(x, y) = x^2 + y^2$$

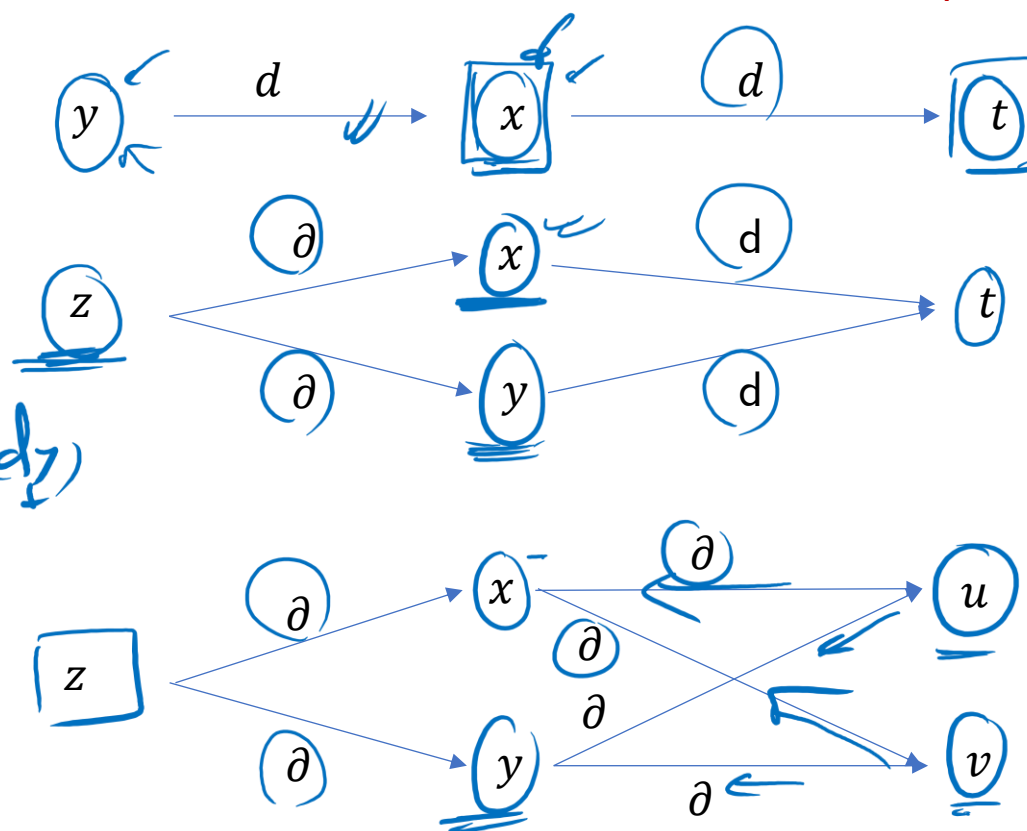
$$y = t^2 \text{ and } x = t^3$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$z = x^3 + y^3$$

$$x = u^2 + v^2 \quad y = 2uv$$

dependent Intermediate Independent



$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$dx = \frac{\partial x}{\partial u} du + \frac{\partial x}{\partial v} dv$$

$$dy = \frac{\partial y}{\partial u} du + \frac{\partial y}{\partial v} dv$$

Gradient of a scalar function

- Consider a scalar valued function maps to a real number $f: \mathbb{R}^n \rightarrow \mathbb{R}$. When we represent the partial derivatives along x and y in vector form, it's called the **gradient vector** $\nabla z: \mathbb{R}^n \rightarrow \mathbb{R}^n$

$$\underline{z} = f(x, y)$$

$$\underline{z} = f(x, y)$$

$$\underline{\nabla z} = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{pmatrix}$$

$$f: (\mathbb{R}^2) \rightarrow (\mathbb{R})$$

(gradient vector)

∇ → nabla
grad' z

- Gradient of a scalar function $z = f(x, y)$ produces a vector ∇z

Exercise : Compute the gradient of $w = (x^2 + y^2)$ and show that the gradient at $(x_0 = 1, y_0 = 2)$, is perpendicular to the level curve at that point.

Jacobian (Gradient of a vector function)

► A vector valued function maps to a vector $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

$$\text{Greeks} = \begin{pmatrix} \text{delta} \\ \text{vega} \end{pmatrix}$$

$$w = \begin{pmatrix} x^2 \\ \ln x \end{pmatrix}$$

$$\underline{w} = \underline{f}(x, y) = \begin{pmatrix} f_1(x, y) \\ f_2(x, y) \end{pmatrix} = \begin{pmatrix} 2xy \\ x \ln y \end{pmatrix}$$

(vector valued function)

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

The Jacobian $\underline{J}$ of $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ will be a $(m \times n)$ matrix where $\underline{J}_{i,j} = \frac{\partial f_i}{\partial x_j}$. (i^{th} output w.r.t j^{th} input).
In the example above, Jacobian is given as

$$\begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

$$\underline{J}_f = \begin{pmatrix} \nabla f_1 \\ \nabla f_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} = \begin{pmatrix} 2y \\ \ln y \end{pmatrix} \begin{pmatrix} 2x \\ x/y \end{pmatrix}$$

Hessian

- The **Hessian Matrix** is a square matrix of 2nd order partial derivatives of a scalar valued function. It's given by the *Jacobian of the transpose of gradient*.

f_x

$$(\nabla_Z)^T = \begin{pmatrix} f_x \\ f_y \end{pmatrix} \rightarrow \text{gradient vector.}$$

$$\mathcal{H}_f = \mathcal{J}_{\nabla_Z^T} = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix} \quad (\text{Hessian matrix})$$

- For example, if $w = f(x, y) = x^2y$, then

$$\nabla w = \begin{pmatrix} 2xy \\ x^2 \end{pmatrix}$$

$$\mathcal{H}_f = \mathcal{J}_{\nabla w} = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}$$

$$\nabla w = (2xy \quad x^2)$$

$$\text{and } \mathcal{H}_f = \begin{pmatrix} 2y & 2x \\ 2x & 0 \end{pmatrix}$$

Quadratic Approximation of a Multivariable Function

- The Taylor series approximation till 2nd order (Quadratic Approximation) for a scalar valued function can be written in a more compact way

$$f(\mathbf{x}) = \underbrace{f(\mathbf{x}_0)}_{\text{(Const.)}} + \underbrace{\nabla f(\mathbf{x}_0) \cdot (\mathbf{x} - \mathbf{x}_0)}_{\text{(Linear)}} + \frac{1}{2} \underbrace{(\mathbf{x} - \mathbf{x}_0)^T \mathcal{H}_f(\mathbf{x}_0) (\mathbf{x} - \mathbf{x}_0)}_{\text{(Quadratic)}}$$

(x,y)
(Const.)
(Linear)
(Quadratic)

$\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{x}$
 $\Delta f = f(\mathbf{x}) - f(\mathbf{x}_0)$

Exercise – Find the quadratic approximation of $f(\mathbf{x}) = \underline{x}e^y$ at $\mathbf{x}_0 = (0 \quad 0)$

Exercises



1. if $U(x, y, z) = 2x^2 - yz + xz^2$, where $x = 2 \sin t$, $y = t^2 - t + 1$, $z = 3e^{-t}$, show that

$$\frac{dU}{dt} \text{ at } t = 0 \text{ is } 24.$$

2. Given $w = f(x, y)$; $x = r \cos \theta$, $y = r \sin \theta$, show that

$$\left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2 = \left(\frac{\partial w}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2$$